MTH 631

Show (0,0) is critical point of the function $P(x,y) = x^2 - 2xy + y^2 - x^4 - y^4$ and it has +ve semidefinite second differentiable at (0,0)

Sol:-

$$P(x,y) = x^2 - 2xy + y^2 - x^4 - y^4$$

Put (0,0)

$P(0,0) = 0$ Hence (0,0) is Critical point of $P(x,y)$

AS,

It will be positive semidefinite when,

$$P(x,y) \geq 0$$

And as $p(x,y) = 0$ Here

So, Proved

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Intereted ingral I1,i2 ..10 mqrks

Ik nested wala

Ik log x integrabal chk kary

Ik x ki piwer q p greater 0 diverage

Ik ds find karnay wala que tha

Ik chaucy sharwaz wala que mcqs kuch file kuch mid

mth631
iterated intrgral definition of
i1.i2
prove s=Rn
boht are of 10marks
objective just 20marks
short
dirichilst test
absolutly integrable
discontinious prove
find df
define continious

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Q.no.1 Find upper and Lower Integrals in $R^n$

For a partition $P = \{R_1, R_2, R_3 \ldots, R_k\}$ of $R$. (10)

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Q.no.2 Evaluate

$$\lim_{x,y \to (2,1)} \text{ where } f(x,y) = \frac{\sin(x-y)}{\sqrt{x-y}}$$

Q.no.3 Define Absolute Integral

Dirichlet's test (5)

Q.no.4 Determine Integral

$$\int_1^\infty \frac{\log x + \sin x}{\sqrt{x}}$$ converges / diverges.

Q.no.5 Calculate $\frac{\partial f(a)}{\partial \phi}$ Where $f(x,y,z) = \log(1+x+y+z)$

$\phi(0,1,0)$ (5 marks)

Q.no.6 Define ~~Point~~ Diameter of a Set

if $S = \{(x,y,z), |x| \leq 2, |y| \leq 1, |z-2| \leq 2\}$

Find $d(S)$



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MTH 631 (MCQ) Final 2019

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(1) A sequence of point $X_r$ in $R^n$ converges to the limit $\bar{X}$ if

$$\lim_{r\to\infty} |X_r - \bar{X}| = 0 \quad \text{OR} \quad \lim_{r\to\infty} X_r = \bar{X}$$

(2) Disconected if S can not be describes as $S = A \cup B$ where $A \neq \phi$, $B \neq \phi$, $\bar{A} \cap B = \phi$ and $A \cap \bar{B} = \phi$.

(3) An open set S in $R^n$ is connected iff it is polygonally Connected.

(4) $$h(x,y) = \frac{\sin u(x,y)}{u(x,y)}$$

(5) $$\lim_{r\to 0} \frac{\sin r}{r} = 1$$

(6) If $\lim_{x\to x_0} f(x)$ exisit then it is unique.

(7) $$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = 1/2.$$

(8) $\lim_{x\to x_0} (f+g)(x) = L_1 + L_2$

(9) $\lim_{x\to x_0} (f-g)(x) = L_1 - L_2$

(10) $\lim_{x\to x_0} (fg)(x) = L_1 L_2$

(11) $\lim_{x\to x} \left(\frac{f}{g}\right)(x) = L_1/L_2$

(12) $\lim_{x\to x_0} f(x) = -\infty$ $\quad 0 < |x - x_0| < \delta.$

(13) $\lim_{x\to x_0} -f(x) = \infty$

(14) $\lim f(x) = L$ (finite) if unbounded.

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Q1. If $S_1$ has zero content and $S_2 \subset S_1$ then $S_2$ has zero content.

Sol:- Let, $S_1 \subset \mathbb{R}^2$ has zero content and $S_2 \subset S_1$

$\therefore$ $S_1$ has zero content.

There is the collection of $\mathbb{R}_i$ such that $S_2 \subset S_1 \subset \cup_i \mathbb{R}_i$

So, $S_2 \subset \cup_i \mathbb{R}_i$ Hence proved.

$\longleftrightarrow$

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→ Determine whether the integral $\int_0^{\infty} \frac{1+\cos x}{(1+x)\sqrt{x}}\,dx$ converges or diverges.

→ Show $f_x$ and $f_y$ exist for all $(x,y)$ but not continuous on the line $y=x$ where $f(x,y) = \begin{cases} (x-y)^2 \sin\frac{1}{x-y} & x \neq y \\ 0 & x = y \end{cases}$

→ If R is degenerate then Prove by using the definition of Riemann integral that $\int_R f(x)\,dA = 0$ if f is bounded on R.

→ Show (0,0) is critical point of the function $P(x,y) = x^2 - 2xy + y^2 - x^4 y^4$ and it has +ve semidefinite second differentials at (0,0).

→ Define intermediate value theorem and Supremum and infimum.

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